Partial Fractions and Improper Integrals

Partial fraction decomposition: distinct linear factors

1 Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.

2 Evaluate $\int \frac{3x+5}{x^2+x-2}\,dx$ using partial fractions (note $x^2+x-2=(x+2)(x-1)$).

3 Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6=(x-3)(x+2)$).

Partial fractions with a repeated linear factor

4 Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.

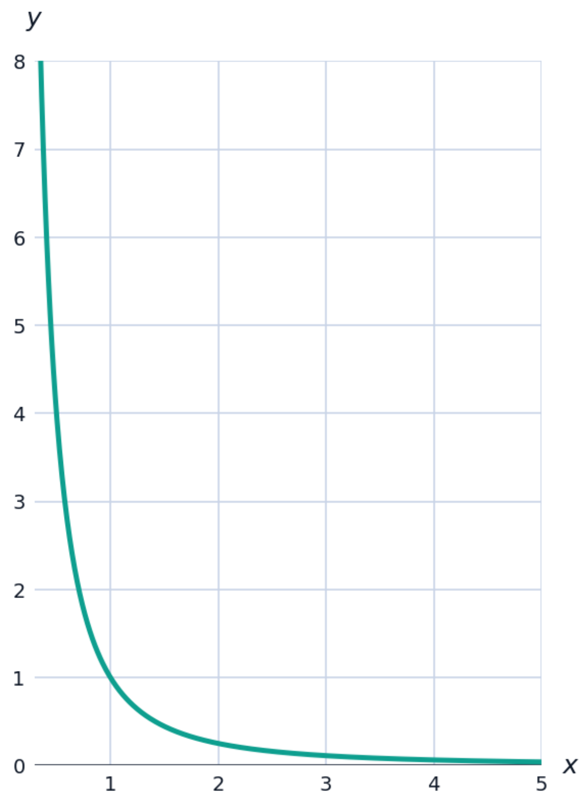
5 Evaluate $\int \frac{2x+3}{(x-1)^2}\,dx$ using the decomposition found above.

Partial fractions with an irreducible quadratic factor

6 Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.

Improper integrals: infinite limits

7 Evaluate $\int_1^{\infty} \frac{1}{x^2}\,dx$, or show it diverges.



8 Evaluate $\int_1^{\infty} \frac{1}{x}\,dx$, or show it diverges.

9 Evaluate $\int_0^{\infty} e^{-2x}\,dx$, or show it diverges.

10 Evaluate $\int_{-\infty}^{0} e^{x}\,dx$, or show it diverges.

Improper integrals: discontinuous integrands

11 Evaluate $\int_0^{1} \frac{1}{\sqrt{x}}\,dx$, or show it diverges (note the integrand is undefined at $x = 0$).

12 Evaluate $\int_0^{1} \frac{1}{x}\,dx$, or show it diverges.

13 Evaluate $\int_0^{2} \frac{1}{(x-2)^2}\,dx$, or show it diverges (the integrand is undefined at $x = 2$).

The p-test for improper integrals

14 State the p-test for $\int_1^{\infty} \frac{1}{x^p}\,dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

15 State the analogous p-test for $\int_0^1 \frac{1}{x^p}\,dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.

Combining partial fractions with improper integrals

16 Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$ using partial fractions.

Three distinct linear factors

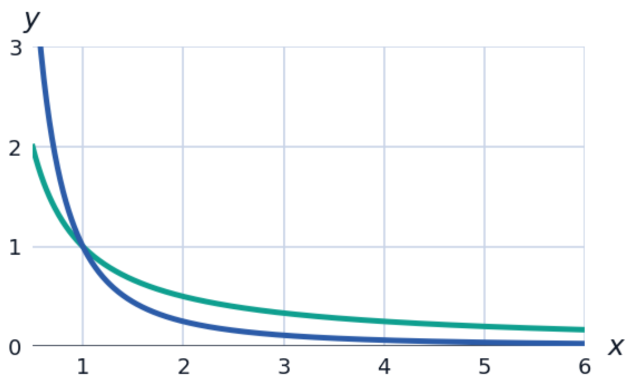
17 Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.

18 Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)}\,dx$ using the decomposition above.

Visualizing convergence: area under the curve

19 The convergent improper integral $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \geq 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.

20 By contrast, $\int_1^\infty \frac{1}{x}\,dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.



Improper integrals with both an infinite bound and a discontinuity

21 Evaluate $\int_0^\infty \frac{1}{\sqrt{x}(1+x)}\,dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.

Comparison test for improper integrals

22 Use a direct comparison to argue that $\int_1^{\infty} \frac{1}{x^2+1}\, dx$ converges, without computing it exactly.

23 Use a direct comparison to argue that $\int_1^{\infty} \frac{2+\sin x}{x}\, dx$ diverges.